

More Than Meets the Eye: Misconduct and Decoupling Against Blockchain for Supply Chain Transparency

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Abstract

The fashion industry has consistently ranked high in terms of its association with modern slavery. While scandals like the Rana Plaza incident have focused the world's attention on Asia, in Brazil, over 35,000 people have been rescued from conditions analogous to slavery in the past 15 years. Outsourcing and the underlying social structures that blur the implementation of outsourcing are central to this problem. Fashion supply chains are highly fragmented and labour intensive, exhibiting power asymmetry and informality. Although supply chain research has focused on how focal firms can control or improve supplier practices, this study examines focal firm misconduct and decoupling. Our research presents an intervention aimed at developing a blockchain solution to create a census of fashion working conditions in Brazil. The project included two non-governmental organisations, two fashion brands and their suppliers, the fashion retail association, a blockchain start-up and the research team (18 organisations). We contribute to both theory and practice, revealing that the blockchain's potential to ease transaction costs is outweighed by governance costs related to third-party supervision. We show that focal firms justify unfair purchasing practices via victimisation and diverting attention from themselves to consumers ('unwilling to pay') and suppliers ('not behaving as expected'), unveiling how these drivers of organisational misconduct lead to supplier decoupling (means–end) and focal firm decoupling (policy–practice). Such underlying social structures sustain inaction or, at best, advance focal firms' visibility of their supply chains while offering no true transparency to the broader society.

Keywords

Modern slavery, supply chain transparency, decoupling, organisational misconduct, blockchain

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1 Introduction

Modern slavery increased by 10% from 2016 to 2021 and now affects over 49 million people, with over 17 million in the private economy (International Labor Organization et al., 2022). Fashion, which employs one in six people on the planet, is among the top-ranked industries. It is a low-technology and labour-intensive field that offers low entry barriers to new and informal suppliers while hosting modern slavery-related issues (Zorzini et al., 2015). In the fashion industry, outsourcing sewing in garment production is a common practice for reducing cost and transparency (Perry et al., 2014). However, outsourcing along supply chain tiers often leads to informal and hidden contracting schemes (Nath et al., 2020; Soundararajan et al., 2021).

Major scandals in the fashion industry such as the Rana Plaza incident have focused the world's attention on Asia (Wang, 2020), but modern slavery persists across the Global

South as well. In Brazil, for example, infringements related to modern slavery remain frequent. Brazil is a country that contains all levels of the fashion supply network – fibre cultivation, weaving, manufacturing and retailing – and this industry generates around 90 billion US dollars annually (top five globally¹). This economic vigour distracts attention from the fact that outsourcing involves blurred boundaries between focal firms and suppliers who are dependent on them but do

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not have appropriate contracts. This blurred liability hinders law enforcement, which is crucial for controlling modern slavery. While European legislation on supply chain due diligence has evolved consistently, in less privileged countries, regulatory progression has been less steady. In the context of Brazil, modern slavery legislation has experienced setbacks during the 2016–2022 period. The most important tool, ‘The Slavery Dirty List’ [free translation], was attacked by corporate lobbyists, who called for its discontinuation in 2016, a matter that escalated to the Supreme Court in 2020.² A recent study mapped 208 legal processes heard by the Brazilian Superior Labour Court (2013–2018) associating modern slavery conditions to fashion outsourcing. The study showed that penalty fees were often issued but unpaid. In 2018 alone, 30 firms were fined the equivalent of 3.5 million US dollars, but only 0.3 % of this amount had been paid by 2020 (Marques et al., 2021). In recent years, the numbers have worsened. According to the International Labor Organization, over 35,000 people had been rescued from modern slavery in Brazil during the 2000–2015 period, but 600 of them ended up in similar conditions at least a second time (International Labor Organization, 2018). In São Paulo, Brazil’s largest city, a 200 % increase in modern slavery was recorded in 2021, when the COVID-19 pandemic served as an excuse for a slowdown on monitoring.³

Supply chain transparency and monitoring are keys to fighting modern slavery. Although major scandals tend to attract societal and academic attention, modern slavery persists as a gradual, continuous threat that often finds a home in the multitiered nature of supply chains (Busse et al., 2016; Tachizawa and Wong, 2015). Since the early 1990s, supply chain transparency has slowly evolved based on supplier audits (Stevenson and Cole, 2018). Focal firms may either engage in monitoring themselves and/or resort to third parties to achieve audit capillarity, yet problem detection at the sub-supplier level is limited (Benstead et al., 2018; Nath et al., 2020). Audit schemes tend to work based on annual local visits. Unique traits of the Brazil context include the low traceability of buyer–supplier contracts, spread informality and lenient legislation. In fact, Brazilian legislation favours outsourcing and small-scale sewing cells with no more than 30 workers each. Hence, suppliers often split production across multiple sewing cells to benefit from tax advantages. Such complexity and dispersion make audit schemes ineffective, providing fertile conditions for modern slavery that are often vexed, dynamic and partially hidden (Christ et al., 2020). However, there is scant research on the failure of audit schemes or alternative solutions for overcoming this issue.

As transparency starts with traceability, blockchain technology (henceforth blockchain) has emerged as a holy grail (Mithas et al., 2022; Sunar and Swaminathan, 2022). Blockchain research began in 2008 in information systems scholarship, and publication output exploded in 2016 (Queiroz et al., 2019). By 2018, the first theoretical studies and reviews had hit the operations and supply chain management (OSCM) field. Treiblmaier (2018) and Saberi et al. (2018) proposed

transaction costs as a lens to understand how the blockchain could reshape supply chains. However, empirical work has only recently been published, and applications are mostly led by start-ups (see: Benstead et al., 2022; Danese et al., 2021). Most importantly, critical research on the trade-offs of blockchain implementation is even scarcer (Murray et al., 2021).

Our study begins by asking the following research question (RQ1): *To what extent can blockchain reduce transaction costs while improving supply chain transparency?*

However, blockchain initiatives to improve supply chain transparency will likely face resistance. Such resistance could relate to the emerging technology itself (Sodhi et al., 2022) or to the consequences of higher supply chain transparency (Chen et al., 2022). One particular form of organisational resistance to new policies that has been increasingly discussed in the OSCM field is decoupling (Nath et al., 2020). Decoupling can occur between policy and practice, for example, between those who set the policy and those who manage daily practice. As long as the policy pleases external stakeholders, what the organisation does internally actually goes untouched (Barratt and Choi, 2007). Decoupling can also be expressed as a gap between means and ends, where policies are indeed implemented but are only weakly related to daily practices (Bromley and Powell, 2012). Initiatives that improve transparency in complex and hidden supply chains may face both forms of decoupling as a reaction (Wijen, 2014).

Modern slavery is ‘not a single, one-off emergency but rather a gradual, continuous threat’, which ‘coupled with the hidden and complex nature of’ supply chains may lead to inaction (Stevenson, 2022, p. 133). The Global South institutional context further favours decoupling. However, previous research has only focused on supplier decoupling (Huq and Stevenson, 2018; Nath et al., 2020; Silvestre, 2015). Our research expands the scope of the analysis to both focal firms’ and suppliers’ complementary forms of decoupling – and, most importantly, the reasons behind such dynamics.

We therefore complement RQ1 with RQ2: *Why does decoupling emerge as a reaction to supply chain transparency against modern slavery?*

Our study was conducted as intervention-based research (IBR; Hasle and Vang, 2021), using the fashion supply network as the level of analysis. The intervention included fashion brands and their supply chains, a blockchain start-up and non-governmental organisations (NGOs), in line with calls for interactions with less traditional stakeholders (Marques, 2019). By expanding the transaction costs lens from the dyad to network level, we found that the blockchain governance costs outweigh the expected savings in local audits. Elaborating on the drivers of organisational misconduct and decoupling, we mapped how multitiered supply chains may actually be used to conceal focal firm misconduct (Kim et al., 2019), helping fashion supply chain actors buffer practices to protect themselves from exposure and outside inspection (Bromley and Powell, 2012). As a result, decoupling was exposed both

at the supplier (means–end decoupling) and at the focal firm (policy–practice decoupling).

The Global South offers opportunities to research unstable macroeconomic conditions and unique social issues (Aguinis et al., 2020; Özkazanç-Pan, 2008). OSCM scholarship has so far emphasised the misbehaviour of Global South suppliers as the key cause of unsustainability (see: Asif, 2020; Villena, 2018) while proposing that Global North focal firms are the ones responsible for changing the status quo (see: Gold et al., 2013; Villena and Gioia, 2018). The predominant assumptions have ascribed power and autonomy to focal firms while blaming weaker partners in the Global South for risks related to greenwashing or modern slavery conditions (Carmagnac et al., 2022; Glover and Touboullic, 2020). Yet, (North)buyer–(South)supplier relationships are often disguised as collaborative, even though they are actually detrimental to the supplier side (Chen et al., 2022). Moreover, focal firm collaboration through third-party certification can ‘fracture over time’, therefore ‘eroding’ trust in joint auditing efforts (Benstead et al., 2018, p. 2307). The failure of supplier audit schemes has often been used an excuse to support the victimisation of focal firms (Parmigiani and Rivera-Santos, 2015; Silvestre, 2015; Wu and Jia, 2018). Thus, our work offers a critique of such dynamics, exposing focal firm liability for perpetuating unfair purchasing practices that lead to decoupling and modern slavery risk.

In the remainder of this article, we discuss the theoretical background, followed by the methodology and the elements of IBR. Next, we present the findings and discussion, including five propositions. Finally, conclusions are provided along with future research directions.

2 Theoretical Background

2.1 Transaction Costs in Supply Chain Transparency

Transaction costs include the costs of negotiating, implementing and monitoring a focal firm’s supply contracts (Emery and Marques, 2011). Transaction costs are driven by bounded rationality and opportunism. The former assumes that cognitive competence is a limited resource. Thus, transactions that demand significant cognitive capacity will tend to be costly, and contracts will be incomplete (Williamson, 2008). The latter assumes that economic actors pursuing self-interest may engage in incomplete or distorted information disclosure (McIvor, 2009). Bounded rationality and opportunism thus increase transaction costs, which are barriers to market governance. As noted by Lumineau and Oliveira (2020, p. 74), ‘if individuals never engaged in opportunistic behaviour, then the market would suffice for the mediation of all transactions’. Therefore, focal firms tend to be attracted by alternatives, such as those offered by new technology, which can alleviate transaction costs.

Opportunistic behaviour and the associated transaction costs that are incurred to prevent it from happening have been widely discussed. Moreover, transaction cost theory has

become a top-of-mind lens in OSCM scholarship (Touboullic and Walker, 2015b), which has expanded on the notions of transaction costs from the dyad to discussing challenges related to multitier supply chains (Meinlschmidt et al., 2018). In the Global South context, research has emphasised institutional voids, that is, the lack of legislation enforcement, mock compliance from suppliers and focal firms’ disregard for supplier violations, all pointing towards supplier opportunism (Huq and Stevenson, 2018). In 2020, Lumineau and Oliveira (2020) called for an increased focus the study of opportunism and for future research on new technologies (e.g., blockchain). The authors also invited research on empirical contexts beyond the US and Europe. The more complex the supply chain, the higher the transaction costs will be to control suppliers and the higher the information asymmetry, which prevents the exposure of organisational misconduct. Information asymmetry thus hinders public scrutiny and supports misconduct (Marshall et al., 2016).

Fashion supply chains are particularly salient regarding misconduct for two main reasons. First, misconduct is particularly hard to trace, given its complexity and dispersion, and modern slavery often does not occur within the focal firms’ direct purview but rather beyond the first-tier suppliers (Busse et al., 2016; Nath et al., 2020; Tachizawa and Wong, 2015). Further, multi-sourcing is a common practice in the fashion industry, and such strategies have been posited as a way to conceal buyer misconduct (Kim et al., 2019). Multi-sourcing accompanied by multitiered structures thus concentrate the labour force in labour-intensive (tier-2) sewing cells,⁴ also called *sweatshops* to denote modern slavery. In these sewing cells, which are mostly located in low-cost countries, informal arrangements prevail and fragile regulation limits social control. Ultimately, supply network complexity may result in a strain between the means and ends, which may be used as justifications for inaction (Caruana et al., 2020).

The second reason fashion supply chains are rife with misconduct is the difficulty in defining how unacceptable working conditions must be to constitute modern slavery. Modern slavery involves workers’ physical and mental conditions at work (Christ et al., 2020). Even when freedom of movement may be preserved, precarious working conditions may constitute modern slavery (Kahya, 2007; Wiengarten et al., 2021). Appendix 1 presents the associated terminologies (see: *e-companion*).

Improving supply chain transparency to trace instances of less-extreme versions of poor working conditions in supply chains can reveal darker versions of forced labour, which are the most difficult to expose. Tracing modern slavery within multitiered structures is a particular challenge (Busse et al., 2016). The transaction costs of third-party audits are high, even for large focal firms that resort to cooperation and resource sharing in audit schemes (Benstead et al., 2021). However, some large focal firms have developed their own standards based on the argument that there is no *one-size-*

fits-all solution. Consequently, the lack of standardisation and lack of industry-level action are obstacles in the fight against modern slavery (Fudge and McCann, 2015).

2.2 Blockchain as a Solution to Supply Chain Transparency

In recent years, blockchain has been promoted as a tool for supply chain transparency at a systemic level. Since 2018, OSCM scholarship has argued that it can be used to enact systemic solutions to sustainability while easing transaction costs (Saber et al., 2018; Treiblmaier, 2018).

A blockchain is a distributed system for capturing and storing a consistent, immutable linear event log of transactions (ledger) between networked actors (Queiroz et al., 2019). There are certain key principles that define a blockchain. First, the information control is decentralised: every piece of information is broadcast to all other nodes, meaning that there is no need for a central authority to approve transactions. Second, the identity behind transactions is protected, and authentication is conducted by nodes of the network, providing a combination of transparency and confidentiality. Third, once a transaction is included in the blockchain, it becomes almost impossible to alter the ledger. Fourth, it is possible to write programs, that is, smart contracts, which define what needs to be checked for a transaction to be confirmed and registered (Tapscott and Tapscott, 2016).

First adopted by the financial markets, blockchain had been discussed in other industries as well, such as pharmaceutical products, containers, wood and food. Blockchain applications can also include real-time monitoring supported by technologies like radio frequency, global positioning systems and internet of things, all of which allow tracking goods in real time and knowing who is performing what in various sectors (e.g., container or supplier monitoring) (Kshetri, 2018). Blockchain applications can use smart contracts to reduce friction (i.e., transaction costs) and support peer-to-peer and collective bargaining (Tapscott and Tapscott, 2016).

Further, it has been posited that blockchain can ease transaction costs (Saber et al., 2018) or even eliminating such costs completely (Tapscott and Tapscott, 2016). For example, blockchain applications can serve as a reputation system that replaces fraudulent rating systems, thus allowing firms to establish the trustworthiness of suppliers (Efanov and Roschin, 2018). Studies suggest that blockchain can not only reduce the effort and the number of intermediaries but also the time needed to monitor suppliers. In addition, smart contracts can increase the level of automation in the signing of buyer-supplier contracts (Shermin, 2017). Beyond transaction costs, it has been argued that blockchain can address core challenges studied in OSCM research: diminishing the bullwhip effect, improving integration and offering scalability to logistics monitoring, all of these particularly useful for multi-tier structures (Agrawal et al., 2021). For example, it might help in avoiding situations like the infamous Chipotle food

contamination case, where the lack of ability to monitor suppliers in real time made it impossible to prevent the breaching of quality standards (Kshetri, 2018).

Ultimately, blockchain could be the key to scalable supply chain transparency. Supply chain transparency can include data about employees clocking in and out of a factory to gauge whether the factory is complying with labour regulations (Stevenson, 2022). Blockchain has been presented as the twenty-first century's 'shop-lifting camera', which combines three elements: traceability, decentralisation of data collection/validation, and breach-proof confidentiality (Shermin, 2017). Such technology is well suited to complex supply networks. However, there few empirical studies have addressed the impact of blockchain on social issues in the supply chain (Pournader et al., 2019). Thus, questions remain: Could blockchains replace traditional monitoring schemes (Greve et al., 2010)? Could this technology reduce misconduct in supply chains in a frictionless manner (low transaction costs)?

In contrast to the overwhelming positive appraisals of blockchain, recent studies by Murray et al. (2021) and Pun et al. (2021) have focused on governance and implementation costs, which are neglected by most researchers. Governance costs are related to data security issues and the 'oracle problem'. Data security costs reflect additional controls for emergent blockchain solutions before achieving scale/trust of all involved actors. Fears related to data security give rise to pre-emptive efforts to avoid errors resulting from removing human discretion. When blockchain usage leads to a dispute between two parties, the need for third-party arbitration may lead to extra costs. Oracle costs refer to impartial third-party experts who are hired to design smart contracts and/or assess data quality (Murray et al., 2021). Thus, such governance costs, most salient during implementation, offer trade-offs to transaction costs. Ignoring these costs may lead to misjudgements in cost analyses (Pun et al., 2021) and disappointment when inflated expectations are not met (Sodhi et al., 2022).

2.3 Decoupling as a Reaction to Supply Chain Transparency

Besides misjudging unforeseen costs, it is likely that blockchain-based initiatives for improving supply chain transparency may face resistance. For example, firms may resist the disclosure of their supply chains (Doorey, 2011), the emerging technologies and the change that they provoke (Sodhi et al., 2022), and the greater fairness that may result from transparency (Chen et al., 2022). One particular form of organisational resistance to new policies increasingly discussed in the OSCM field is decoupling (Nath et al., 2020).

The foundations of decoupling date back to the 1970s. Weick (1976, p. 4) stated that 'intentions are a poor guide for action, intentions often follow rather than precede action, and that intentions and action are loosely coupled'. Meyer and Rowan (1977) framed decoupling as a thoughtful disconnection between organisational structures aiming at legitimacy

and daily organisational practices. However, only recently has OSCM scholarship begun to consider decoupling (Barratt and Choi, 2007), and most recent research has investigated decoupling in suppliers' response to pressures related to sustainability (Huq and Stevenson, 2018; Nath et al., 2020). We centre our analysis on decoupling while expanding the scope to the interplay between focal firms' and suppliers' complementary forms of decoupling.

Decoupling has been conceptualised at two levels: policy–practice and means–ends. It is valuable to distinguish between these two forms because of their varied consequences for organisations (Bromley and Powell, 2012). The policy–practice gap exposes the misalignment between policy and practice over time. Policy–practice decoupling allows organisations to adopt conflicting policies in response to external pressures without disturbing their daily operations (Bromley and Powell, 2012). Often, decoupling occurs across the organisational hierarchy: Those in charge of policy please external stakeholders, while those in charge of daily activities continue to perform untouched (Barratt and Choi, 2007). Such isolation in daily activities has been framed as 'buffering' (Bromley and Powell, 2012), that is, as long as policy adoption alone builds legitimacy, real change can be delayed. Arguments for delaying action include a lack of capacity and the need for time and costs to adapt (Nath et al., 2020).

In means–end decoupling, policies are implemented that have a weak relationship with the intended outcomes (Asif, 2020). Organisations are inherently complex. Within that complexity, organisational participants may consciously recognise that some actions are of limited utility but still pursue them to please some stakeholders (Bromley and Powell, 2012). The allocation of resources between practices that only aim at legitimacy and core practices disperses energy. A lack of transparency increases the potential of dissociation between means and ends and eventually renders the implementation of sustainability standards impossible (Wijen, 2014).

Focal firms that prioritise their reputation will often transfer responsibility for issues related to modern slavery to their suppliers (Meehan and Pinnington, 2021). In other words, they define policy, but action is 'outsourced' to their supply chain, thereby overburdening suppliers. This is probably why previous research has focused on supplier decoupling, specifically as means–end decoupling. Suppliers may engage in a concealment strategy, attempting to prevent nonconformity with sustainability practices by deliberately hiding their violations. Alternatively, suppliers may engage in a consensual strategy that entails attempting to implement partial sustainability practices by negotiating with other actors (Nath et al., 2020). In fact, when focal firms 'outsource' an action to their supply chain, they are also engaging in decoupling, but this issue has been less addressed in the literature.

Decoupling (in both forms) often occurs due to information asymmetry between firms and their stakeholders

(Huq and Stevenson, 2018). Supply chain transparency may reduce information asymmetry (Chen et al., 2022). In addition, blockchain solutions can ease the transaction costs of implementing supply chain transparency (Saberli et al., 2018; Treiblmaier, 2018). However, if resistance arises from supply chain actors (in the form of decoupling), blockchains may fail to support supply chain transparency, resulting in the persistence of modern slavery risks. Our intervention aims to understand the extent to which blockchain can ease transaction costs and enhance supply chain transparency (RQ1) and why transparency to mitigate modern slavery is met with resistance (RQ2).

3 Methodology

3.1 Intervention-Based Research

Recent OSCM research has urged scholars to eschew the 'quintessential ivory tower syndrome' (van Aken et al., 2016, p. 1) and critically engage with the problems they study (Touboulie et al., 2020). The former calls for effectiveness over explanation, designing/testing over theorisation, and pragmatic validity over theoretical validity (van Aken et al., 2016). The latter invites engagement over observation, exploring researcher agency and embeddedness over time to allow access to what is usually inaccessible, as well as critical thinking, reflexivity, and contextualisation 'by deprivileging the criterion of generalisability' (Touboulie et al., 2020, p. 10).

IBR is a methodological umbrella that can help answer those calls. IBR can aim to solve a socio-technical problem in a given context while reflecting on the theoretical implications of such endeavours (Fendt and Kaminska-Labbé, 2011; Iivari and Venable, 2009). Although historically marginalised, IBR has received growing attention in recent years, resulting in guidance for researchers (Chandrasekaran et al., 2020). Having considered different guidelines for IBR, this study follows Oliva (2019, p. 710), seeking to 'explore the role that interventions (the ultimate manifestation of design science) can play in testing and developing theory (the ultimate goal of an academic endeavour)'. According to this framework, IBR is structured into two modes of enquiry. Mode 1 is grounded in action research and can test and expand theories (Checkland, 1985), while mode 2 holds explanatory power grounded in a process perspective, which can lead to theory development (Langley, 1999). In other words, an intervention can explore *why the chosen theory cannot explain the phenomenon?* (mode 1) and *why the system behaves in an unexpected manner?* (mode 2).

Along our journey, unexpected results allowed us to explore both modes of enquiry. In Appendix 2 (*see: e-companion*), the iceberg analogy is used to show that mode 1 occurred at the *surface*, that is, investigating blockchain and transaction costs (RQ1), whereas mode 2 delved deeper into the *bottom of the iceberg*, exploring the decoupling and misconduct that led to interruption of the intervention (RQ2).

Mode 1 reflects action research, which favours practical utility and encourages change (i.e., a successful intervention) while also yielding insights for theory elaboration. The action researcher simultaneously takes action and creates knowledge (Touboullic and Walker, 2015a). The practical utility of this study was the significant involvement in the development of a mobile app to survey working conditions. Yet, midway through the intervention, we recognised the misfit between the adopted theory and our data, as resistance prevented the blockchain from advancing. Unpacking audit and governance costs enabled a granular revision of the total costs involved. In the Section 4, we provide evidence of why the theory was insufficient to fully inform the intervention and how it failed to anticipate specific developments. In Oliva's (2019) terms, we reflected on situation S to inform why theory T was not capable of explaining the intervention.

After identifying the misfit, the research team turned its attention to mode 2 or why the system behaved unexpectedly. The focus slowly shifted from action research to process research, where theory elaboration is central and practical utility is secondary. Although we were still involved in the project report and relationship with other stakeholders, our attention was increasingly oriented towards explaining why the new situation S' differed from the expected situation S*. In process research, abductive reasoning is utilised to combine the *inside-out* perspective of transaction costs with the *outside-in* perspective of organisational misconduct (Greve et al., 2010) and decoupling (Bromley and Powell, 2012). We had entered the project aware of the high unpredictability inherent to social and technical dimensions of the intervention, but we were quite surprised by the results.

3.2 Intervention Management and Data Collection

This study involves IBR focused on a particular industry in which modern slavery risks are salient, the fashion industry, and a particular country, Brazil, a representative of Latin America and the Global South. The Brazilian fashion supply chain can be fully local, from farm to store, and yet exhibits high levels of opacity. Fashion brands normally outsource their production to tier-1 suppliers, who take charge of buying fabric, cutting fabric, and finishing the product while re-outsourcing the sewing to tier-2 sewing cells. While the conditions at the fashion brands and tier-1 suppliers are usually adequate (often above standards), workers in sub-suppliers often endure precarious conditions. There is higher turnover, and tier-2 suppliers eventually informally outsource production they cannot manage to other sewing cells, thus creating informal/illegal tier-3 suppliers. The further away the brands are from the consumer, the higher the risk of modern slavery. Multitiered outsourcing often portrays suppliers in a dependent relationship with focal firms as intermittent (non-dependent) to shield those focal firms from problems when modern slavery is uncovered. Such blurred liability has been overcome in some cases in which employee testimonials,

contracts, and invoices were used as evidence, but cases that reach the court and lead to convictions represent only a tiny fraction of the reality (Marques et al., 2021).

The goal of the intervention was to develop and pilot test a mobile app that could invite workers to participate in a monthly census of their working conditions. A key step of IBR is researcher–client agreement. Unlike a non-participatory approach, action research entails real interaction between the researcher and the loci of the research (Iivari and Venable, 2009). This project was led by the research team after an offer of funding from a prominent NGO in the industry, here referred to as NGO-A, in December 2017. NGO-A was founded by the owners of a major European fashion retail chain that has been operating in Brazil since the 1980s. NGO-A is dedicated to promoting transparency and systemic change by funding projects that aim to change the fashion supply chain towards sustainability and transparency. Engagement with NGO-A automatically brought Brand-A into the project to provide us with access to fieldwork. The research team and NGO-A identified and invited a blockchain start-up focused on developing blockchain solutions oriented towards scalable social change. Together, the research team and the start-up constituted the project's Core Team.

The research project spanned from December 2017 to March 2020. In December 2017, negotiations between Researcher-A and the NGO-A, and the proposal was developed and signed in 2018. In January 2019, Researcher-B joined the team, and the intervention lasted until March 2020, when the expansion from the pilot to full implementation was interrupted. Researcher-A was the project manager throughout the full period. Researcher-A, Researcher-B, or both were present at all data collection points, except after Researcher-C joined the team in October 2019 and conducted the last round of interviews.

The project plan consisted of involving two pilot supply chains, each composed of one brand, two tier-1 suppliers, and four tier-2 suppliers. In 2018, a second brand took part in the research but dropped out in the first month of project execution. Thus, the project kicked off in February 2019 with only Supply Chain-A, while searching for a second brand (a strict requirement of NGO-A). By June 2019, NGO-B and Brand-B joined the project and provided some extra funding. NGO-B is a small-sized organisation but is recognised for its work towards sustainability. Brand-B is a Brazilian brand founded in the 1980s, with a much smaller scale and a price ticket 7–9 times higher than that of Brand-A. Therefore, the brands did not view themselves as competitors. Supply Chain-B was formed by one focal firm, two tier-1 suppliers, and three (instead of four) tier-2 suppliers. Non-disclosure agreements (NDAs) were signed by all firms in both supply chains.

Although both brands were leaders in sustainability and transparency initiatives in the country, there were also valuable differences between the two supply chains: different origins (Brand-A is European and Brand-B is local), different sizes

(while Brand-A has one of the largest sales turnover in the country, Brand-B has less than 10 % of Brand-A's sales), different target consumers (Brand-A is a fast fashion chain, while Brand-B is a premium brand), and different sourcing geography (Brand-A's local suppliers are concentrated in the São Paulo region, while Brand-B's suppliers are located in Rio de Janeiro and Southern Brazil). These characteristics supported purposeful sampling, thereby enhancing the external validity of the research findings.

Appendix 3 (*see: e-companion*) presents a map of the intervention and data collection instances. The left side of the figure shows the supply network organised in three layers: the institutional layer with the government, the research team, and the start-up; the pre-competitive layer, including the Fashion Retail Association (henceforth 'Association', which represents one-third of fashion sales in Brazil) and NGOs that engage with multiple fashion brands; and finally, the supply chain layer, which represents workers, suppliers, and the fashion brands. The figure presents both the 18 organisations directly involved in the intervention and entities that were not interviewed but mentioned repeatedly during the study, such as the government. The right side of the figure quantifies the data collection: meetings, workshops, interviews, and fieldwork tours, including in loco recorded interviews.

The data collection was carried out using different methods that can be clustered in three groups, varying in frequency and formality. The first cluster refers to the core team (the research team and the Start-up), which interacted through multiple e-mails, phone calls, conference calls, and meetings in São Paulo. The second cluster covers the two brands and the two NGOs, in which most of the interactions were concentrated on the five group workshops conducted over one year, although phone calls and e-mails were also registered for analysis. The third cluster comprises the supply chain actors (tier-1 and tier-2 suppliers and their workers), where interactions were restricted to the formally scheduled conference calls and local visits and the two interviews with the Association director.

Researcher-A, Researcher-C, NGO-B, and Brand-B were based in Rio de Janeiro, whereas NGO-A, Brand-A, the Start-up, the Association and Researcher-B were based in São Paulo (note: hired to benefit from proximity). There were 34 meetings, 25 of which took place via conference calls and nine in a rented facility in São Paulo. The same facility hosted five group workshops that served as the key milestones of the project (Appendix 4, e-companion).

Interaction with suppliers and their workers was carried out through two rounds of fieldwork visits spread from São Paulo (Supply Chain-A) to the south of the country (Supply Chain-B), totalling 20 field visits. The field trips included 15 semi-structured and recorded interviews and multiple unrecorded conversations. In addition, during the second tour, 60 workers provided (unrecorded) feedback about the mobile app in a series of meetings conducted by 2–3 Core Team members with groups of 5–7 workers in a round circle. In addition, two

interviews were conducted with the Association in São Paulo, the first midway during the project and the second during the project closure. In the last stage of the project, Researcher-C conducted three interviews with NGO-A's manager and the start-up team.

Appendix 5 provides a summary of the intervention timeline (27 months), organised into four phases ranging from 3 to 12 months each. For each phase, the figure indicates the core activities and the resulting data collection. The diagram also indicates when each researcher joined the project. The four phases not only depict the passage of time but also mark key shifts in data analysis and theory elaboration. Appendix 6 offers the full list of interviewees (*see: e-companion*).

3.3 Data Analysis and Visualisation

The data analysis adhered to the recommendations of Chandrasekaran et al. (2020). First, we developed a rich description of our intervention and included appropriate data. Second, the theory from which we departed was a plausible starting point, that is, transaction costs. Third, the insight into the need to recalibrate and develop theory further represented sufficiently new and interesting knowledge.

We employed a multitude of data visualisation techniques to convey the message and portray both the process perspective in its phases and the emerging concepts and their relationships. In the Section 4, power quotes are complemented with vignettes at the beginning and end of the section.

Vignettes have recently been utilised by some OSCM researchers to study a women's empowerment programme in Ghana (McCarthy, 2017) and a slavery detection initiative in Southeast Asia (Benstead et al., 2021). In both studies, the authors advanced our understanding of supply chains from a Global South perspective, using vignettes to visualise fieldwork reflexivity. In our study, the vignettes were written in the first person, drawing heavily on Researcher-A's diary notes and reflections, as reflexivity is central to action research (Checkland, 1985). An action research project begins with an action aimed at producing change and ends with a reflection on the learnings. Therefore, first-person vignettes can help embed the reader in the data analysis and reflections.

While vignettes can help with immersion, data coding supports theoretical propositions. To organise the data and produce the analysis, we employed the temporal bracketing strategy (Langley, 1999). Bracketing data in phases helped describe the processes and mechanisms identified. For this, 'evidence is (was) drawn together to examine how the context affects these processes, and what the consequences of these processes are on the future context and other relevant variables of interest' (Langley, 1999, p. 703). Organising the intervention in phases also helped to reveal mechanisms that only surfaced once the intervention advanced from inception to development and validation. Temporal bracketing helped us move from messiness to a framework without misrepresenting the inherent complexity of the phenomenon.

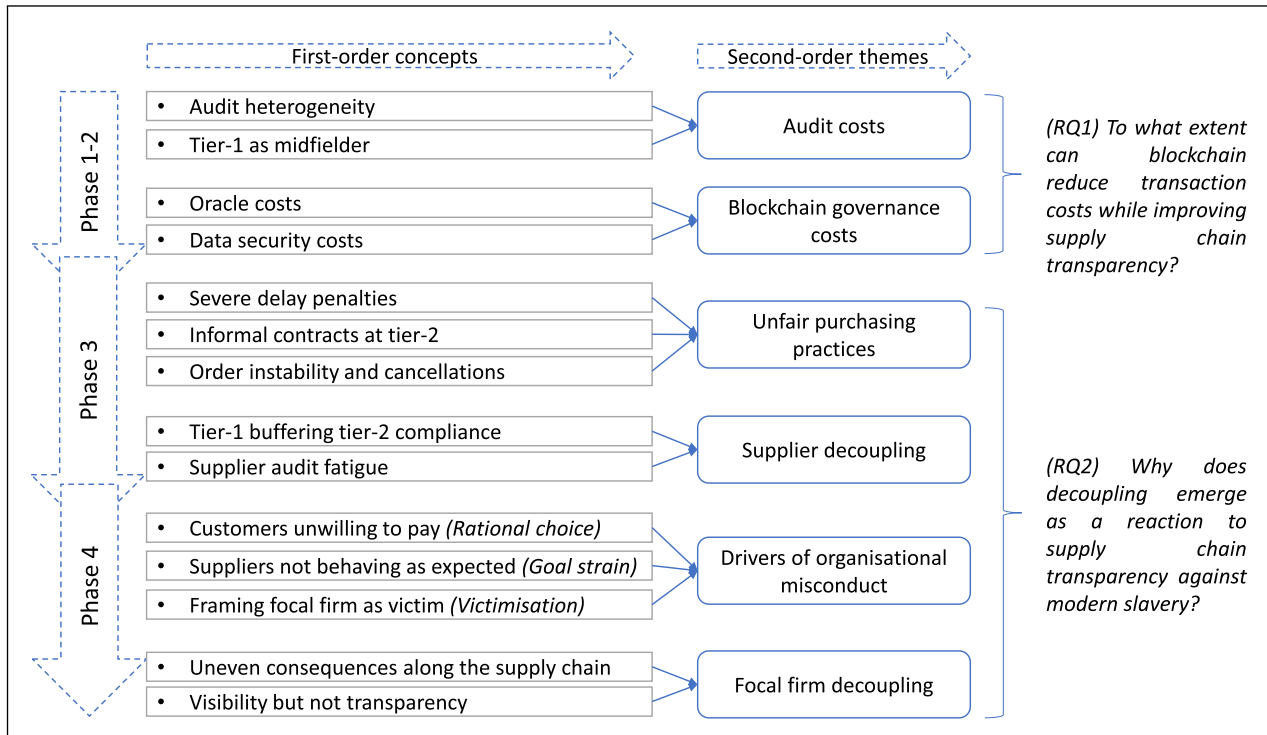


Figure 1. Data structure.

According to Langley (1999), process studies should balance accuracy, generality, and simplicity. Temporal bracketing offers conditions for excellent accuracy (as long as the phases depict time shifts properly), with moderate generality (single-case limitations), and moderate simplicity (given the challenges in portraying a process view and a variable view). Most importantly, the chosen strategy focused on uncovering the driving mechanisms of the phenomenon under study.

The data coding followed the two-cycle approach proposed by Saldaña (2013). Within each temporal bracket, the first cycle was focused on reviewing the corpus and building a foundation based on *holistic coding*, which is appropriate for studies with a wide variety of data forms, such as transcripts and field notes. This type of coding is applicable when the researcher already has an idea of what to look for. We started with transaction costs and then considered the reasons why the intervention failed to mitigate such costs. The first-order concepts were grounded on power quotes extracted from the recorded interviews and complemented by e-mails, field notes, and other data sources, which were key to supporting the transition from first-order concepts to second-order themes. The second coding cycle aimed to synthesise and conceptualise the first-order concepts, and we explored *focused coding* to map the most frequent and significant codes to develop salient categories (Saldaña, 2013). Figure 1 presents the data structure based on first-order and second-order stages, illustrating the phases when concepts emerged as well as which RQ they address more directly.

Additional data is presented in Appendix 7 – representative power quotes for first-order concepts and Appendix 8 – a sample of mobile app screenshots translated into English (see: *e-companion*). While preserving confidentiality, this evidence offers further insights on the intervention.

3.4 Reflections on Research Quality

An interventionist must be conscious that interventions seek to build on artificial systems that do not yet exist (i.e., ‘will it work?’) and contrast with the natural sciences’ explanation of existing natural systems (i.e., ‘is it valid or true?’) (Oliva, 2019). An interventionist must also ensure the development of a knowledge database that can support both practical and academic objectives (Nair et al., 2011), respecting both the ‘researcher–client agreement’ and their professional duty to contribute to academia. Finally, an interventionist must state their instance, perspective, and possible limitations. Researcher-A had extensive experience in the fashion industry prior to academia, while Researcher-B and Researcher-C had no prior experience in the industry. All researchers were born and raised in Brazil, which shaped their worldview but also helped in developing stronger relationships with the participants.

Team diversity enriched our discussions and provided a source of triangulation, which also occurred while confronting the different perspectives formed during the project phases. Further, the study also included the triangulation of methods and data to strengthen its reliability and internal

validity (Stake, 1995). Method triangulation involved the combination of participant observation and in-depth interviews, semi-structured interviews, and document analysis. For data triangulation, interviews were compiled with the same interviewee at different points in time, allowing observation of the same phenomenon at different locations. Finally, a post-project review was performed with key participants, who were asked to verify the data analysis and correspondence to reality. This step included three interviews conducted by Researcher-C and a final online meeting with NGO-A and Brand-A, all of which were conducted in early 2020.

4 Findings

This section is structured according to the four phases described in Appendix 5 to highlight temporal bracketing. In addition, we frame the first and last phases (1 and 4) around first-person vignettes to embed the reader in the data analysis and reflections. Conversely, we have organised the intermediate phases (2 and 3) around raw quotes, with the intention of providing direct access to raw data and nuances of speeches across workers, suppliers, brands, NGOs, and association representatives. In other words, we offer a journey that starts and ends with the narrator's voice (Researcher-A), with the original voices presented between.

4.1 Phase 1. Project Setup (and Behind-the-Scenes Access)

On December 22, 2017, Researcher-A received a compliment from an old friend on social media. She had moved from Rio de Janeiro to São Paulo a few years before, and they had lost contact. She had become a manager at NGO-A, which funds projects to improve supply chain transparency. A first meeting via Skype happened on January 31, 2018, and as a result, Researcher-A joined a project to conduct an intervention to test the extent to which blockchain could help improve supply chain transparency. After attending a couple of meetings in May 2018, Researcher-A became the project manager and received an initial grant to develop a 12-month project for 2019 (while travelling between Rio de Janeiro and São Paulo). In the following months, the project team decided to focus solely on working conditions and modern slavery and visited one tier-1 and one tier-2 supplier of Brand-A.

Vignette 1: The gut punch of reality behind the scenes.

July 17, 2018. Today, I visited a 'pilot supply chain', starting with Brand-A's headquarters, followed by a direct supplier and a tier-2 supplier, in São Paulo, the fourth largest city in the world with 22 million people. Brand-A and tier-1 suppliers are both located in the same industrial cluster, one hour from my hotel. The tier-2 supplier is based on a poor neighbourhood almost two hours away from the industrial cluster. This is a supply chain in a big city with long distances and too much traffic.

The visit to the tier-1 supplier is a pleasant surprise. Clearly a top supplier, the owner (and general manager) takes me around the facility and shows me the production control team, their software, and the visual management tools that support order management and quality control of 20 tier-2 suppliers spread across São Paulo with whom they work. They also have an audit team that works on four wheels running across these tier-2 suppliers, like a milk-run system. They work hard to achieve high quality and productivity and to offer a design that matches with what their customers want – and they happen to be one of the five largest fashion retail chains in the country. The owner is also proud to state that he is a 'recurrent volunteer' for Brand-A's improvement projects and is happy to help with the 12-month project we are about to launch.

The visit to the tier-2 supplier is a 'gut punch'. The supplier is situated in a region of the city where the postal service does not deliver, taxi drivers refuse to go, and Uber prohibits their drivers from going due to violence in the neighbourhood. We get there by dropping from an Uber ride eight blocks away in front of a barber shop. There, we meet the owner (let's call her Ana) to get a ride in her car to the facility. Ana works diligently to make a living, and she explains that this barber shop is where she can, for example, get something delivered by Amazon when needed. We enter this neighbourhood 'excluded from São Paulo map' to see a sewing cell (tier-2 supplier) that works for the tier-1 suppliers of some of the major fashion brands in the country.

Ana had started small, and growth had come intrinsically linked to family. Since sewing cells should not grow too much (approx. 30 workers) to take advantage of the tax benefits, Ana has opened multiple sewing cells that work like production cells. Two of them are managed by her daughters, and the third one is managed by her former husband turned business partner. As the four sewing cells supply to tier-1 suppliers of major fashion brands, this means not only monthly surveillance from the tier-1 suppliers but also annual audits from each fashion brand, and on top of that, the annual audit of the Brazilian Fashion Retail Association Program. These audits overlap but also have differences in their requirements; thus, compliance with all demands multiplies the required efforts and investments. Ana tells me that while Brand-A audit emphasises security (i.e., electric installations, maintenance, fire alarm), another major fashion brand has stricter controls on 'quarteirização' (informal tier-3 sewing cells), and a third one is very much focused on internal working conditions (i.e., lunch time, paid leave, etc.). Ana complains about the increasing costs, continuous priority shifts, and inclusions in the audit checklist. With each visit comes a new surprise and a new requirement.

When asked about her relationship with her employees, she described herself as a mother to them. Some need guidance on their personal finances. Some are beaten by their husbands and ask for early leave to prepare dinner on time. Ana changed her tone when talking about her fear of having an illegal immigrant and how to avoid that. Surprisingly, she also told the story of

an auditor who had acted as a stalker, following her employees outside work to probe into the working conditions and ask personal questions. Overall, we feel her struggle to cope with increasing controls, low margins, and a lack of demand stability from her customers.

I leave the visit and take the plane back to Rio de Janeiro, lost in thought about the world we have created for ourselves...the fashion supply chain clearly involves a huge imbalance between major fashion brands that reach around one billion dollars in annual turnover, tier-1 suppliers that are capable of achieving a turnover of a couple of million dollars in annual revenue (working with approximately 300 employees and managing 20 sewing cells), and finally, the tiny scale of these tier-2 suppliers, with no more than 30 employees and striving to survive another month. Within this unbalanced and dispersed supply chain lies a horde of workers, mostly female, and their voices are the only ones capable of uncovering the truth about fashion working conditions.

If only we could let them speak using technology to streamline communication and offer the transparency of their voices to the final consumer, we could then ease the multiple audit schemes and seamlessly produce trust.

By September 2018, we had presented a proposal to the boards of both Brand-A and NGO-A and received a green light for the project. By October, we had hired Researcher-B. By December, we had to change our blockchain partner from a multinational to a local start-up, and the second brand had cancelled its participation before the project could start. Despite these surprises, we had agreed to kick off with Brand-A alone while searching for a second brand. By February 2019, the project had a detailed governance structure and time schedule, and its budget was approved. At this stage, expectations were quite high. There was an overall excitement about the blockchain's potential to contribute to fighting modern slavery in the fashion industry. Concerns were concentrated on ensuring that the solution would help in achieving standardisation, reducing the transaction costs related to multiple (and ineffective) local audits, and putting ourselves in the shoes of the workers, that is, accounting for resistance, access to technology, technology literacy, and so forth.

4.2 Phase 2. Design Thinking (And Transaction Cost Trade-Offs)

During this phase, we analysed the Association's audit scheme and decided that it should serve as the basis for the project's definition of modern slavery. This audit scheme classified issues related to working conditions that were in line with the literature, as described in Appendix 9 (*see: e-companion*).

Although the Association's political strength seemed to result in standardisation, with its army of auditors covering thousands of local audits every year across the country, there was still a long way to go. The heterogeneity of audit schemes was still high, as pictured in the previous phase. At this stage, we inferred that the blockchain-led solution could be

the final push towards standardisation, as technology favours automation, and automation favours the adoption of common standards. In addition, by producing a solution adherent to the Association's audit scheme, the chances of attracting other fashion brands would be higher. By avoiding the specificities of the current firms, the solution would then appeal to the entire industry. We also needed wider participation to cover rising governance costs.

In six months, the project advanced through workshops and the first fieldwork tour to Supply Chain-A in February 2019. In March 2019, we held the first group workshop. In April, we signed a contract with NGO-B and Brand-B. In June, we held a second group workshop with both. During this phase, we adopted a design-thinking approach to capture the needs of involved stakeholders, including the focal firms, tier-1 suppliers, tier-2 suppliers, and workers, with the results summarised in a social canvas.

We had entered the fieldwork highly concerned about possible resistance to a blockchain solution. Initially, the Core Team was mostly concerned with the workers' resistance to using a mobile app and responding to a monthly survey. Therefore, significant emphasis was placed on testing the workers' familiarity with mobile phones, their perceptions of the mobile app, their reaction to the survey's writing style, and how clear the questions were to them. There were many discussions on how to probe on sensitive issues related to payment terms and benefits, work hours, work environment and fatigue, as well as concerns about how to handle responses to open-ended questions:

'I think we need to be very responsible with the information we are going to collect. It is not just about what we are going to ask directly but about the risks if we leave an open question. And if someone writes something like "help me" – anything that puts us in the situation of needing to take someone out of danger – are we prepared for that? Is there a way for the app to capture this? And if we cannot handle this, how should we proceed?' (Director NGO-B)

In contrast to our concerns and those expressed by the NGOs, during our fieldwork, the workers expressed surprising excitement with the project:

'It is the first time in 27 years in this profession that someone asked me how I feel'. (Worker T2A22)

'I felt embraced. This way, we will have a voice'. (Worker T2A11)

Interestingly, while NGOs and fashion brands voiced concerns about workers' resistance, they saw no risks regarding the suppliers. Their discourse indicated a healthy relationship with their supply base, regardless of the fact that the suppliers selected for the pilot project were among the best (thus, it

would be natural to expect them to gladly collaborate). Nevertheless, the fieldwork revealed unease on the part of suppliers regarding the lack of standardisation of the audit schemes. Suppliers complained that they needed to work with multiple fashion brands to ease demand instability and seasonality while also having to comply with different audit schemes. For example, each scheme could frame the same problem as minor or major. As a result, one fashion brand might demand immediate action, while another would shift its attention to a different issue, accumulating requests for adaptation.

An example of this is immigrant work. Manager T1A1 was emphatic in telling us that his Paraguayan employee was always the one to be requested for an interview during audits from one fashion brand, while another fashion brand would leave this aside and focus on capacity management to avoid *quarteirização* (when tier-2 suppliers outsource to informal tier-3 suppliers). If this were true, the project could reduce friction and audit burdens. Therefore, besides addressing the pains of each stakeholder in the supply chain, we also aimed to produce a solution strongly grounded in the Association's audit scheme to increase homogeneity and ease supplier adoption.

By the end of this phase, there was a mix of excitement with what the blockchain could provide and concerns over the emerging governance costs. To what extent could something as complex as blockchain technology be incorporated as an easy-to-use solution for shopfloor workers? Clearly, the technology angle of the socio-technical problem outweighed the tensions related to governance and transparency that started to surface.

4.3 Phase 3. Pilot Testing (And Decoupling as Resistance to Transparency)

In the following six months, the start-up developed the beta version of the mobile app survey. We conducted another group workshop with both focal firms to capture their feedback, and in parallel we planned and executed the second fieldwork tour, with the goal of testing the mobile app survey with 300–400 workers across all suppliers participating in the project.

By the time we started scheduling the second fieldwork, the growing pressure of the audit schemes had affected our project. The suppliers found it inconsistent that brands kept pressuring them for lower prices and tighter deadlines while increasing auditing demands. Audit schemes were voluntary for brands but compulsory for suppliers once the brands they supplied opted in. We soon realised that our project was considered part of the growing audit workload. At this stage, the consequences of the unfair purchasing practices that had become the norm in the industry began to emerge.

From the suppliers' perspective, the demand variation imposed by the fashion brands (when postponing and cancelling orders without prior notice) was a driver of working condition instability, which could in turn increase the chances of workers responding incorrectly to the survey. Suppliers complained about fashion brands' purchasing policies: cancelling purchasing orders quite frequently and on short notice

while establishing harsh penalties for any delays: *'What cannot happen is a late delivery. If you delay delivery, you pay a 1 % penalty fee. If the delay accounts for more than 5 % of the total, the penalty fee increases to 5 %'* (Manager T1A1).

Suppliers could not offer job security to workers without demand stability. Moreover, they emphasised that such unfair purchasing practices were spread across the industry. Tier-1 suppliers felt squeezed between price pressure and rigorous audit schemes. Once again, immigrant work was a salient issue. Bolivians were described as more productive than Brazilians but harder to tame into audit requirements:

'Before fashion brands started (to care about) social responsibility, audits, all those things, I had 26 sewing cells with Bolivians. One day, I approached them to say the business would change. Only five accepted and 21 have left'. [...] 'The fact is that Bolivians sew like no one else, which impacted my efficiency. While they can produce 10 difficult pieces, Brazilians can produce only one. However, their health standards do not go according to the audit schemes'. (Manager T1A1)

Focal firm pressure on tier-1 suppliers often led to tier-1 pressure on tier-2 suppliers, cascading unfair purchasing practices to the weaker links in the supply chain. Even the tier-1 suppliers that avoided *'letting their sewing cells down'* eventually had to cascade postponements/cancellations. It became obvious that the lack of long-term contracts was, in fact, a common practice. Suppliers felt that they were constantly *'in the dark'*, *'holding the bull by the horn'*, and facing financial struggles: *'Last year, I had no service for a full week. (...) But then I still need to pay the workers to stay at home. And that is the problem. There's no way I can handle this. Last year we stayed in the red [negative bank account] the full year because of that one week'* (Owner T2A11).

With little demand instability, tier-2 suppliers bounced from one customer to the other, and as a result, informal contracts were almost the norm: *'Actually, nobody has a contract. It is all under word-of-mouth assurance. If you provide a good service, you keep supplying'* (Owner T2A11). Frequently, instability led to lay-offs: *'Last year I had almost 30 workers, but five of them I had to... [making a gesture at her throat to indicate lay-off without saying the word]'* (Owner T2A12). Demand instability thus had devastating consequences on workers' job security. Although this was clearly the reality of the industry, the focal firms seemed to refrain from acknowledging it.

As our specific intervention progressed, frustration turned into resistance to our own initiative. Suppliers resisted further participation, as they saw each visit as an extra burden. In fact, it became clear that the focal firms were not adequately informing the suppliers about the intervention goals, durations, and benefits. The focal firms simply assumed that the suppliers would participate, 'outsourcing' the role of engaging suppliers to researchers who had no power or incentives to offer. It felt as if the focal firms considered both tier-1

and tier-2 suppliers almost as their own property, while the owners of the sewing cells obviously had a very different view. Despite the workers' receptivity, the suppliers themselves saw no benefit and only an additional burden. We were supposed to survey 30 workers from each supplier during the second fieldwork tour, but we only visited half of the suppliers, and we collected feedback from fewer than 10 workers at each facility. As a result, instead of 360 responses, we were left with 60 feedback points, most of them informal and poorly registered due to time pressure to conclude the visit.

In fact, although we aimed at developing a socio-technical solution by improving quality/frequency while reducing transaction costs, we found that the very nature of purchasing practices induced supplier instability and eventually could elicit modern slavery. Most importantly, supplier resistance sabotaged the project. Surprisingly, the focal firms seemed to wish to enhance traceability without reconsidering their own purchasing practices and their impact on supplier costs or working conditions. Moreover, social and institutional dynamics seemed to help firms rationalise their reasons for not changing their purchasing practices, which were detrimental to their supply base.

4.4 Phase 4. Project Interruption (And a Lesson About the Status Quo)

Vignette 2: The gut punch of the status quo

January 14, 2020. Today is probably my last trip to São Paulo funded by the project. Two full years have passed since my first contact with NGO-A, approximately one year for the proposal and another year for the intervention. After visiting a candidate global retailer that could be interested in the project, even though they raised multiple restrictions, and after my last phone call with NGO-A and the start-up team at the hotel room, I order a taxi while a feeling of frustration engulf me. A project borne to transform the industry is about to die... or at the very least, evolve into something that I did not dream of when we started this journey...

It became clear that the intervention affected each stakeholder in different ways, raising competing constraints that prevented it from moving further from pilot to full implementation. The truth is that governance costs to please all stakeholders take the investment to a level where industry engagement is needed. NGO-A has always told us that their initial investment was 'seed money' and that fashion brands' buy-in was key to their continued support. At the same time, fashion brands' participation came conditional on a focus on visibility (their own control), thus delaying true transparency. As a result, NGO-A's interest was clearly fading away. NGO-B and Brand-B are small players and have offered us all they had during the pilot stage. They are eager to move ahead, as long as someone else

takes over the financial burden. Last, but not least, the Association has emphasised high heterogeneity across fashion brands. Most are interested only in visibility, and even the front runners seem to be waiting for the others to make the first move. This puts us in a vicious cycle, where everyone is waiting for someone else to 'open the dance floor'. As a result, there is no dancing.

I fly off from São Paulo to Rio de Janeiro not knowing this was not only my last project trip but also my last trip to the fourth largest city in the world for a long time (Covid-19 struck us exactly two months after...)

During the last two months of the project, we had multiple calls for combinatorial analysis with the start-up, NGOs, brands, and the Association. We ended up having more calls in March 2020 (calls on 10 of 22 working days, sometimes more than one per day) than at any other time in the project. Yet, no concrete results came out of them. It became clear that fashion brands were not yet suffering enough pressure to increase transparency, as expressed by one brand manager: '*Have you ever heard the phrase: Sweatshop is wrong unless the shoes are cute?*' (Manager, Brand-A).

The project slowly entered fadeout mode. After March 2020, contacts with industry players revealed that our project had inspired advancements in traceability but with no intent of offering transparency to consumers and stakeholders. As this research project concluded, true fashion supply chain transparency is still non-existent in Brazil (or anywhere else, as far as we are aware).

5 Discussion

5.1 Exposing the True Cost of Blockchain (RQ1)

We begin the discussion by answering RQ1: *To what extent can blockchain reduce transaction costs while improving supply chain transparency?*

The fashion supply chain is dispersed and hard to control. Focal firms often source from hundreds of tier-1 suppliers that each employ dozens of sewing cells (tier-2 suppliers). The further upstream one goes in the supply chain, the more dispersed, and consequently, opaque, it becomes. Growing pressures to improve control and transparency have led to audit schemes that translate into transaction costs. A lack of trust and redundancy of controls (parallel audit schemes) further add to transaction costs. Our intervention revealed that each tier-2 sewing cell was usually visited monthly by its tier-1 suppliers and audited by each focal firm once per year. Further, the national association also ran its own audit scheme, again locally and once per year. As a result, one sewing cell – that supplied to, for example, three focal firms – was likely to be audited 16 times per year. However, the focal firms still lacked control, while differences in audit schemes were overburdening suppliers. In response, tier-1 suppliers often acted

as midfielders, buffering problems from tier-2 sewing cells. Audit failure was clearly occurring.

Previous research has emphasised focal firms' need to embrace coopetition (Benstead et al., 2018) and work through firm associations to improve supply chain transparency (Villena, 2018). In our study, both coopetition and a strong association were in place, but the problem persisted. Given this scenario, blockchains would seem to be a promising solution to ease the problem (Kshetri, 2018; Saberi et al., 2018). Scholars have emphasised the potential of blockchain solutions to automate flows while substituting for trust and adding traceability in a frictionless manner. Thus, they are particularly capable of reducing transaction costs. In the fashion supply network, a blockchain solution could reduce audit costs, as digitalisation would provide standardisation and increase the frequency of control. However, our intervention exposed the need for complex governance that produced unforeseen transaction costs.

Similar to studies on blockchain use to prevent counterfeiting, we identified blockchain costs related to data security (Danese et al., 2021) and trade-offs between implementation and transaction costs (Pun et al., 2021). Such implementation costs took the form of 'oracle costs', that is, third-party experts to oversee blockchain smart contracts on a regular basis (Murray et al., 2021). Such governance costs would thus counterbalance the reduction in local visits, at least in the short to medium term. Therefore, while blockchain might reduce costs related to local audits, unforeseen governance costs/investments could overshadow the expected economy. Blockchain governance would only become viable if implemented on an industry scale, where cooperation could avoid overburdening isolated focal firms. These conclusions led us to our first proposition:

PROPOSITION 1. *Blockchain can save audit costs, but governance costs outweigh such savings without industry scale and commitment.*

Questioning why industry-level implementation did not occur in a scenario with a strong association led us to shift from mode 1 (RQ1) to mode 2 (RQ2) and explore 'evidence for the development of new theories not considered prior to the intervention' (Oliva, 2019, p. 712).

5.2 Unveiling Decoupling and Organisational Misconduct (RQ2)

Beyond facing cost trade-offs, the intervention was hampered by emerging resistance in the form of decoupling. Thus, we now address RQ2: *Why does decoupling emerge as a reaction to supply chain transparency against modern slavery?*

Supplier resistance did not emerge without reason. Suppliers' refusal to take part in the intervention was a consequence of the focal firms' unfair purchasing practices. Such practices included, on the one hand, severe delay penalties for suppliers

and, on the other hand, order instability and cancellations, which led to disruptions in production, urgent requests, and very limited demand predictability (Chen et al., 2022; Meehan and Pinnington, 2021; Muratore and Marques, 2021). Suppliers thus had to live in a state of permanent tension, complying with price pressures and absorbing demand instability while accepting workloads and costs from parallel audit schemes. Redundancy did not ensure control while overburdening suppliers. As the focal firms simultaneously persisted in their unfair purchasing practices and tightened audit schemes, the suppliers felt doubly pressured. Such hypocrisy enabled the focal firms to outsource the risk, responsibility, and costs of modern slavery upstream to suppliers (Glover and Touboulic, 2020), which in turn led to supplier resistance in the form of decoupling.

Supplier behaviour emerged in the form of means-end decoupling. Suppliers found themselves with limited agency to respond to focal firms' impositions (Carmagnac et al., 2022). Since suppliers were economically dependent on focal firms, they responded to increasing audit pressures with means-end decoupling in two ways. First, tier-1 suppliers acted as buffers to tier-2 suppliers' non-compliance. Tier-1 suppliers could not simply cancel tier-2 sewing cells that failed to comply, or they would not meet the cost pressures. Instead, tier-2 suppliers' non-compliance was hidden or at least delayed from exposure to the focal firms.

By outsourcing the burden of modern slavery mitigation to suppliers (Meehan and Pinnington, 2021) while simultaneously sustaining price pressures, focal firms closed their eyes to informal sub-contracting and work instability (Soundararajan et al., 2021), 'creating a dangerous cascade of unfair pay' (Chen et al., 2022, p. 4306). Our second proposition indicates the joint effects of unfair purchasing practices and growing audit demands:

PROPOSITION 2. *The combination of increasing audit costs for suppliers and unfair purchasing practices of focal firms leads to supplier resistance and buffering of non-compliant practices which configures supplier decoupling (means-end).*

Unfair purchasing practices seemed to spread across the industry. Prior research has suggested that coopetition is a driver of transparency (Benstead et al., 2018; Villena, 2018). Here, we observed another facet of coopetition: the legitimisation of unfair purchasing practices. To understand such dynamics, we resorted to drivers of organisational misconduct. Greve et al. (2010) presented a robust review of organisational misconduct, stating that 'misconduct is a behaviour that flourishes where legal, ethical, and social aspects have blurred lines rather than clear limits' (Greve et al., 2010, p. 36). Recently, the OSCM scholarship has employed the lens of misconduct to draw attention to practices that are widespread and persistent, despite their negative consequences for society (Skilton and Bernardes, 2022).

The first driver identified in this study was coded as *rational choice*, which emerged when firms justified delaying costs to prevent misconduct by downplaying their consequences (Greve et al., 2010). In our case, focal firms blamed consumers' unwillingness to pay premiums for transparency as a reason to delay action: 'Have you ever heard the phrase: *Sweatshop is wrong unless the shoes are cute?*' Despite their share of responsibility, consumers were not heard, consulted, or engaged with (Carmagnac et al., 2022). Portraying them as greedy and unaware helped justify focal firms' inaction rather than taking ownership of the responsibility to inform consumers (Chen et al., 2022). Moreover, focal firms used alleged consumer unwillingness to pay and volatility as arguments for passing price pressure and demand instability downstream in their supply chains.

The second identified driver reflected a *strain* (tension), which was observed in the framing of goals as unachievable (Greve et al., 2010; Wijen, 2014). In our case, focal firms diverted attention from their own practices and blamed supplier behaviour. Tier-1 suppliers were framed as the ones responsible for negotiating with tier-2 suppliers, as if price pressures from the brands to tier-1 suppliers would not translate to such pressures to tier-2 suppliers. In addition, tier-2 suppliers were blamed for accepting demand beyond their capacity, thereby fragilising the supply chain. The focal firms reasoned that none of these issues were linked to their own unfair purchasing practices. If the cause was external to the focal firm, supply chain transparency could be framed as unachievable (Doorey, 2011).

The third identified driver reflected a *victimisation culture*. Culture can permit misconduct by socially weighing victims' claims or diverting attention from the original victim (Greve et al., 2010). This notion echoes climate change deniers, for example, when they downplay the impact of carbon emissions on society to justify corporate inaction. Modern slavery research has suggested that the efficiency paradigm supports focal firm victimisation (Caruana et al., 2020). If focal firms could see value in ethical behaviour and transparency, some would advance towards supply chain transparency (Stevenson and Cole, 2018). However, if they jointly shared a common culture of denial, homogeneity would lead to inaction at the industry level.

Together, the drivers of rational choice, goal strain, and victimisation culture supported a discourse to absolve unfair purchasing practices by blaming the others: consumers, suppliers, and fierce competition. Note that, on the one hand, the competitive side of cooptation led to price pressures, and, on the other hand, the cooperative side of cooptation diffused misconduct. The highly ambiguous nature of modern slavery, linked to the lack of transparency, further contributed to inaction (Stevenson, 2022). Suppliers were thus left with no alternatives. While OSCM research has often considered how focal firms can prevent opportunism by suppliers (Asif, 2020; Nath et al., 2020; Villena, 2018; Villena and Gioia, 2018), few studies have investigated focal firm misconduct. Our findings led to a proposition that captures these dynamics:

PROPOSITION 3. *The victimisation of focal firms and diversion of attention to consumers ('unwilling to pay') and suppliers ('not behaving as expected') configure the drivers of organisational misconduct that are used to justify focal firms' unfair purchasing practices.*

Besides supporting unfair purchasing practices, the identified drivers of organisational misconduct also led to focal firm decoupling. While supplier decoupling has been addressed recently (Nath et al., 2020), factors related to focal firm behaviour have been overlooked. In particular, focal firm power in the supply chain should lead to higher responsibility, not the reverse (Asif, 2020). Such a power imbalance became clear in the fashion supply chain, cascading from the brand to tier-1 suppliers, then from tier-1 to tier-2 suppliers, and finally to workers at the end of the supply chain, who are exposed to modern slavery conditions.

Our study revealed that focal firm decoupling took a different form than supplier decoupling. While supplier behaviour was characterised by means–end decoupling, decoupling at the focal firms took the shape of a policy–practice gap. The policy–practice decoupling reflected symbolic adoption. While a transparency policy was highlighted in discourse and corporate reporting (Meehan and Pinnington, 2021), weak implementation led to little or no change in daily routines (Bromley and Powell, 2012).

When the mobile app was developed to allow full supply chain transparency, the focal firms' responses shifted to a stepwise approach. This meant improving the visibility of focal firms to their supply chains without providing transparency for stakeholders, which has been defined as a 'secrecy strategy' (Marshall et al., 2016). From a policy perspective, transparency was a core value for many focal firms in the association. However, their practices involved visibility and control without transparency. As such, the failure to comply resulted in harsh consequences for the suppliers but no consequences for the focal firms. In other words, without supply chain transparency, blockchain could be used to control suppliers, but no link between supplier non-compliance and brands would be made public. Non-compliant suppliers could be excluded from the association list, but no specific links between each brand and these non-complaint suppliers would be made. This served as a 'smokescreen' (Doorey, 2011). Without a clear timeline on how and when to progress from visibility to transparency, visibility became an end in its own right rather than a means to transparency (Asif, 2020).

By transferring the responsibility for modern slavery to supplier practices (Meehan and Pinnington, 2021) and consumer choices (Chen et al., 2022), focal firms could divert attention to weaker links in the supply chain – a hypocritical practice that decouples discourse from practice (Glover and Touboullic, 2020). In our fourth proposition, we elaborate on how focal firms' response to transparency pressures included buffering their supply chain practices and protecting them from exposure/outside inspection (Bromley and Powell, 2012):

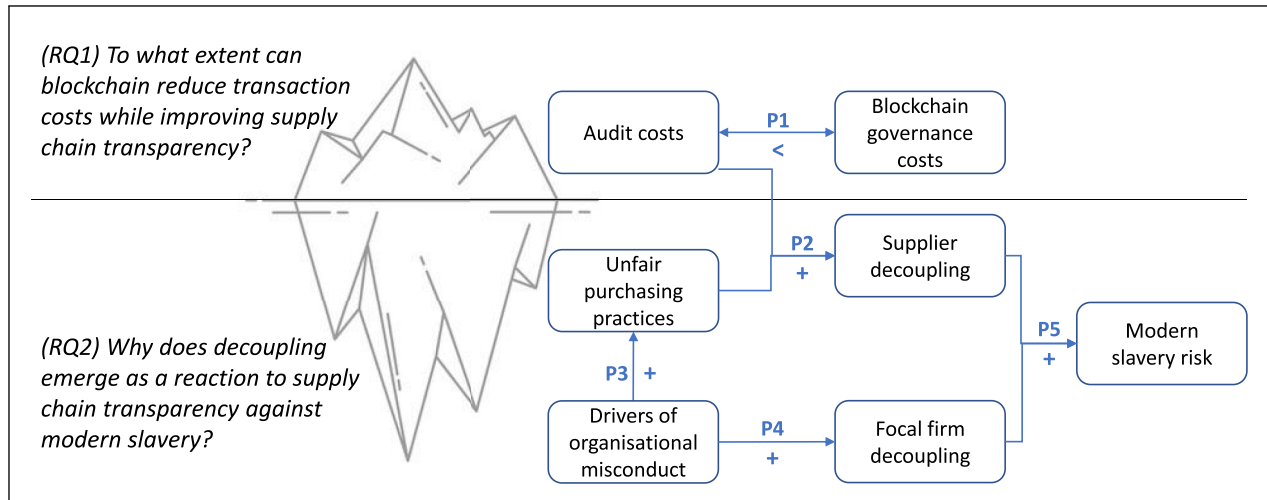


Figure 2. Conceptual framework.

PROPOSITION 4. *The victimisation of focal firms and diversion of attention to consumers ('unwilling to pay') and suppliers ('not behaving as expected') configure the drivers of organisational misconduct that support focal firm decoupling (policy–practice).*

Although the mechanisms depicted in Propositions 1–4 do not ensure modern slavery, they eroded the assurance that modern slavery could be under control. Complexity related to the fact that in isolation, each brand was not to blame. Rather, industry-wide practices produced price pressure and demand instability, which hurt the supply chain, pushing suppliers to engage with informality (Soundararajan et al., 2021) and thus providing fertile soil for modern slavery. However, blurred liability and focal firm practices that eroded margins, intensified pressure on suppliers, and could lead to working condition infringements were less likely to be systemically challenged. As a result, focal firms did not engage in reflections regarding their own commercial practices (Meehan and Pinnington, 2021).

Modern slavery instances tend to spread in the supply chain, worsening during crises, such as COVID-19 (Cole and Shirgholami, 2021), but advancing as a gradual, continuous threat (Stevenson, 2022). In this scenario, supply chain transparency is promised and implied but never truly confirmed (Meehan and Pinnington, 2021). Our final proposition summarises decoupling's impact on the risk of modern slavery:

PROPOSITION 5. *Supplier decoupling (means–ends) and focal firm decoupling (policy–practice) increase the risk of modern slavery.*

In Figure 2, we return to the iceberg analogy, with the five propositions and a conceptual framework.

6 Conclusion

Thus far, little research has focused on the justifications that focal firms provide for not investing in supply chain transparency. The OSCM scholarship will continue to provide an 'unhelpful caricature' of the problem without an in-depth analysis of 'the organisational and supply chain dynamics that give rise to it; and the business actors and models through which it flourishes' (Caruana et al., 2020, p. 252). Our study sheds light on the dynamics that allow organisational misconduct and decoupling to persist, particularly in the Global South context (Huq and Stevenson, 2018; Nath et al., 2020). In this way, we offer a methodological contribution to our field by conducting IBR.

Only extreme cases of opportunism become public. By the same token, most occurrences of misconduct pass unnoticed. In both cases, most incidents are opaque, as (a) perpetrators try to hide them due to social desirability, and (b) potential informants and whistle-blowers often see the costs as higher than the benefits of denouncing. However, there is a key difference between the two: While opportunism looks at wrongdoing due to self-interest, making it an inside-out perspective (Lumineau and Oliveira, 2020), organisational misconduct looks at wrongdoing as socially constructed, therefore making it an outside-in perspective (Greve et al., 2010).

When misconduct is a widespread practice, addressing it requires systemic interventions. Focal firms are the supply chain leader, engaging in explicit coordination and holding the most power. Thus, as much as they can support small suppliers, they can also impose on them disadvantages if misconduct is accepted (Glover and Touboulic, 2020). Although our field has devoted increasing attention to the role of social dynamics (Kim et al., 2011; Marques et al., 2020), there is scant research on how misconduct becomes

widespread in supply chains (Skilton and Bernardes, 2022). We address this based on theory elaboration, and we offer five propositions.

In our first proposition, we argue for the limitations of blockchain solutions in reducing transaction costs. We show that a lack of trust leads to governance costs that outweigh the potential for reducing audit costs (Murray et al., 2021). High governance costs call for cooptation. Our conclusions clash with those of OSCM research that has focused on the benefits of blockchain (Saberli et al., 2018; Treiblmaier, 2018). Such findings have strong implications for practice. Blockchain projects must incorporate governance and implementation costs, which can only be accomplished with industry-level adoption.

We then provide four propositions that unpack the underlying reasons why transaction costs alone cannot explain barriers to supply chain transparency and why the project was hijacked by drivers of organisational misconduct deeply rooted in the sector's culture that allowed decoupling to emerge, perpetuating the risk of modern slavery. The findings inform practice by showing that decoupling should be addressed within focal firms' realms as much as along the supply base. However, we offer guidance not only to practice but also to policy. If industry-level misconduct is not addressed by policy, and malpractice is widespread, no focal firm can be easily blamed in isolation.

Our study resonates with previous research on whistle-blowing. Whistle-blowers have often been framed as illegitimate and have suffered retaliation. Moreover, retaliation has been stronger when whistle-blowers have disclosed information about systemic and deep-seated wrongdoing as opposed to isolated incidents (Kenny, 2018). Our work reveals how organisational misconduct blocks an intervention that would allow workers' voices to be heard and expose well-established detrimental practices towards suppliers and workers.

Could blockchain technology offer a complement to whistle-blowing strategies as a crowd-sourced worker voice? According to Reinecke and Donaghey (2020, p. 28), 'Ideally, worker-driven supply chain governance will involve worker voice across different levels (...) and giving them an active role in monitoring and enforcement ensures accountability towards the intended beneficiaries of decent work initiatives – the workers themselves'. We show that any socio-technical solution must build a common ground to pacify different stakeholders with competing demands (e.g., labour, capital, consumers) if we are to advance to a more transparent and fairer society. Future research should dig further into how technology can complement traditional whistle-blowing strategies to support workers' voices in supply chains (Stevenson, 2022). Yet, technology can only serve fairness and justice if we trigger a shift away from labour as risk towards labour as care and stewardship (Caruana et al., 2020). Without the social dimension, the technical dimension is powerless (Mithas et al., 2022).

We advance OSCM research on modern slavery. Despite the fit between OSCM and modern slavery research, the flow of people in supply chains has been overlooked compared to the flow of goods. Caruana et al. (2020) called for horizontal (vs. vertical), multitier (vs. singular), and bottom-up (vs. top-down) conceptions of supply chain relations. Our work also dives deeply into such relations. We show convergence among focal firms in building a discourse that shifts liability to both ends of the supply chain: 'consumers are unwilling to pay' and 'suppliers are hard to trace'. The endorsement of such discourse, however, is not explicit.

Recent research has discussed the failure of interventions to address modern slavery due to the stickiness of a dominant logic. Hasle and Vang (2021) called for unpacking how this dominant logic obstructs decent work interventions. In our study, we unpack the mechanisms through which the following problem can be answered: 'While buyers typically blame suppliers for decoupling, the problem is in reality related to the fact that both buyers and suppliers pursue strategies based on cost-cutting' (Hasle and Vang, 2021, p. 59). We expose a dominant logic deeply rooted in organisational misconduct that is widespread among focal firms. Focal firms associate instances of modern slavery with supplier misconduct or accidents rather than reflecting on their own practices. They also tend to blame the complexity and costs associated with transparency when explaining their supply chain opacity (Chen et al., 2022). Thus, a victimisation discourse serves as the 'alibi' they need. Future research could look at the conditions that can break this dominant logic and reshape firm behaviour.

Our work reveals the underlying social conditions of modern slavery in the Global South supply chains. The reconceptualisation of the Global South under a postcolonial framework explores the notions of mimicry and hybridity (Özkazanç-Pan, 2008). Mimicry emerges as the attempt of the coloniser to make the colonised copy their culture. Furthermore, to dominate lands and people by way of mimicry, the coloniser needs the colonised nation to be articulated as a homogeneous space. Meanwhile, according to the notion of hybridity, (supply chain) practices emerge through encounters rather than through static, nation-based cultural differences. Future research could utilise a postcolonial lens in considering ways of reframing capital-work relations and supply chain power dynamics that sustain organisational misconduct and hinder workers' voices.

Predominantly, OSCM scholarship seems to echo focal firms' voices and reasoning, positioning misconduct at the feet of suppliers rather than the other way around. If we do not truly change and ensure that suppliers and workers are heard, the behind-the-scenes truth in fashion will never meet the eyes of society.

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Supplemental Material

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Notes

1. Tex Brasil (2020) 'Dados da indústria têxtil e de confecção referentes à 2019', available at: <http://texbrasil.com.br/pt/imprensa/dados-da-industria-texil-e-de-confeccao-em-2015/> (accessed 25 Nov 2020).
2. Among other defeats, in 2019 the Federal Government extinguished the Ministry of Labour and transferred its responsibilities to the Minister of Economy. As a counterbalance, in 2020, the Supreme Court considered the Dirty List constitutional. The dispute continues.
3. Available at: <https://g1.globo.com/sp/sao-paulo/noticia/2022/01/29/numero-de-trabalhadores-resgatados-de-situacoes-de-trabalho-analogo-a-escravidao-na-cidade-de-sp-aumenta-quase-200-percent-em-2021-diz-mpt.ghtml> and <https://www.cnnbrasil.com.br/nacional/pandemia-impoe-retrocesso-ao-enfrentamento-do-trafico-de-pessoas-no-brasil/> (both accessed 25 Feb 2022).
4. We avoid the pejorative term sweatshop and opt for the term 'sewing cell'. We do not use the term 'workshop' also often applied in the fashion industry for tier-2 suppliers, because we make many references to the five workshops conducted as full day activities during the project.

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